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Full Length Article

Integrating spatial-temporal risk maps with candidate trajectory trees for explainable autonomous driving planning

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ABSTRACT

With increasing public concern about autonomous vehicles, there is a growing demand for developing explainable autonomous driving planning technology. Traditional risk field methods use handcrafted potential field models to explain driving risks in a scenario. When explaining highly interactive scenarios, such prior knowledge-based methods still lack flexibility, leading to insufficient interpretability. In this study, we first propose the concept of a risk map that can be seen as a discrete, ego vehicle's view form of the risk field. We then design an explainable trajectory planning framework that integrates risk maps with the candidate trajectory tree generated by trajectory prediction models. We further filter safe candidate trajectories from the tree on the basis of their cumulative risks in the risk maps and then select the optimal trajectory to execute by balancing other driving objectives. The validation results in various real-world scenarios demonstrate that our method can generate understandable risk maps and explain the risk differences between trajectories. Open-loop experiments show our model's advantages in terms of safety and efficiency for the trajectory planning task. An analysis of runtime demonstrated its potential for real-world applications.

1. Introduction

The rapid progress of autonomous driving technology is revolutionizing transportation (Yi et al., 2024; Zhang et al., 2024a, 2024b). As the penetration of autonomous vehicles (AVs) increases, their advantages in safety and efficiency can improve the traffic environment (Huang et al., 2024a, 2024b; Liu et al., 2023; Zhang et al., 2023). Although the future of this technology is promising, its black-box decision-making process has raised public concerns (Hussain and Zeadally, 2018). Owing to the lack of explainability, users do not understand the choices made by AVs and worry about potential failures in unexpected situations (Kim and Kelley-Baker, 2021). This issue further hinders the deployment of autonomous driving in the real world (Koopman, 2024).

A promising approach for addressing the above issues is to develop explainable autonomous driving planning methods (Atakishiyev et al., 2021; Kuznietsov et al., 2024; Omeiza et al., 2021, 2022). Some studies have made progress in this direction. One classic visual explanation is the driving risk field (Wang et al., 2015). As shown in Fig. 1a, researchers have designed potential field models to represent driving risks that align with human cognition. Owing to its quantifiable and continuous form, the risk field can be used for trajectory planning through gradient descent or sampling (Tian et al., 2020; Wang et al., 2024a). However, owing to their limited flexibility, such predefined methods still struggle to generate easily understandable explanations in highly interactive scenarios (Wang et al., 2022). To be precise, they cannot explain the relationship between trajectory planning and driving risk in these scenarios. With the widespread application of question-answering (QA) systems in autonomous vehicles, some studies have designed planning algorithms based on models with reasoning capabilities, such as trees (Gyevnar et al., 2022; Omeiza et al., 2021) and large language models (LLMs) (Dong et al., 2023; Peng et al., 2024; Xu et al., 2023). Although natural language-based explanations generated by trees or LLM can enhance interaction with humans, they usually provide only a qualitative comparison of different driving choices.

In this study, we provide a new perspective for explainable autonomous driving planning. We integrate risk maps with a candidate trajectory tree to explain the relationship between trajectory planning and driving risk. As illustrated in Fig. 1b, a candidate trajectory tree is

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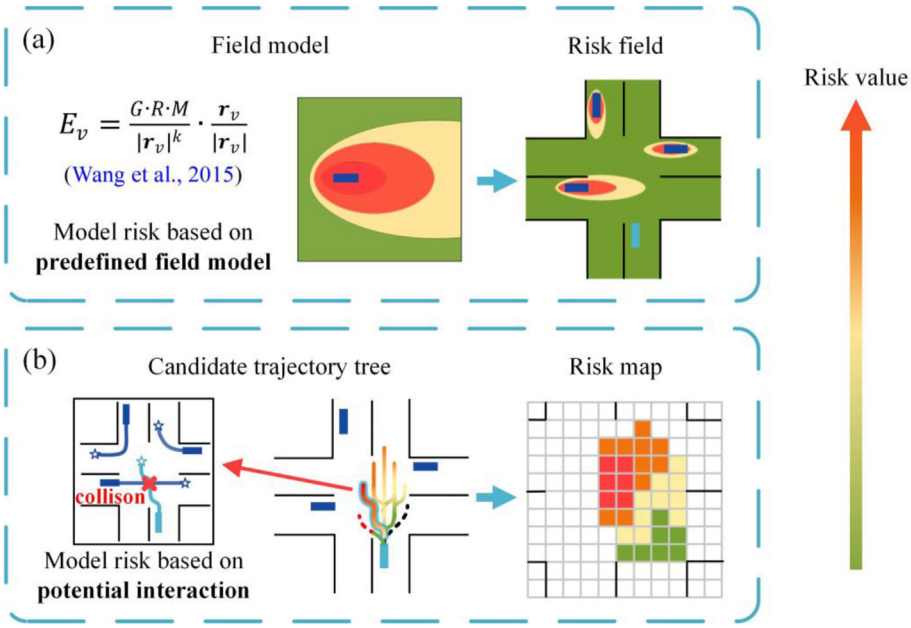

Fig. 1. Comparison of risk modeling between (a) the conventional risk field and (b) the proposed method. The risk field uses predefined field models. Our method models risk on the basis of the potential interaction. In terms of risk representation, the risk field presents risks in a continuous form from a global view, whereas the risk map represents risks in a discrete form from the ego vehicle's view.

constructed to simulate the interactions with surrounding vehicles in the scenario, thereby modeling dynamic driving risks. The tree with risks is then mapped into the so-called risk map, which can be seen as a discrete, ego vehicle's view form of the risk field.

Specifically, we design a behavior tree that includes various driving behaviors (e.g., overtaking, accelerating, and cruising). In the planning stage, we first convert the behavior tree into a candidate trajectory tree and simulate interactions between the ego and its surroundings. When a possible collision is detected, we set the risk to the points of this candidate trajectory branch on the basis of their distance from the collision. Multiple branches in the same subtree average the risk and then propagate it to the parent branch. With the tree and its risk, we further generate visual explanations of the driving risk and then make safe driving choices. We consider prediction uncertainty to set the trajectory width and plot the trajectories onto the grid map. The levels of the tree are mapped onto different maps. Thus, the maps can represent the spatial and temporal evolution of driving risk. On the basis of the spatiotemporal risk maps, we filter safe candidate trajectories from the trajectory tree according to their cumulative risk. Finally, we select the optimal candidate trajectory by considering other objectives, such as efficiency and comfort.

We test our method on real traffic datasets. The test scenarios include intersections, roundabouts, and ramps. The generated risk maps can explain how driving risk changes over time. In addition, the maps can intuitively reflect the risk levels of different driving trajectories. The open-loop simulation experiments show that the proposed algorithm can plan trajectories with lower risk and higher efficiency than human driving. An analysis of the runtime indicates that our method has good computational efficiency. In summary, our contributions are outlined as follows:

1) We propose a new method for driving risk modeling in explainable autonomous driving planning. Instead of prior knowledge-based field models, we use the candidate trajectory tree to infer risks from predicted interactions in a posterior manner, achieving more flexible and precise risk modeling.
2) We design a risk representation method to improve explainability in highly interactive scenarios. By mapping the tree into spatial–temporal risk maps, we can demonstrate the evolution of future risks and explain the relationship between the risk and driving trajectories.
3) In the trajectory planning task, our method outperforms human driving in terms of safety and efficiency. By considering the cumulative risk of the trajectory and other driving objectives, we can select the optimal candidate trajectory from the candidate trajectory tree.

The rest of the paper is organized as follows. Section 2 reviews the related research on explainable autonomous driving planning. Section 3 presents the proposed method. Section 4 introduces the experiments, and Section 5 provides the conclusions.

2. Related works

In this section, we review several mainstream explainable autonomous driving planning algorithms. We introduce three main types of methods: risk field-based, tree-based, and large language/vision language model (LLM/VLM)-based.

Risk field-based: Similar to fields in physics, driving risk fields consist of predefined risk value scalars in space that dynamically change over time (Wang et al., 2015). Risk fields provide a unified visualization framework to explain the impact of various traffic elements on the road environment, such as road constraints (Wang et al., 2015), dynamic obstacles (Wang et al., 2024b), traffic control (Hua et al., 2022), and occlusions (van der Ploeg et al., 2024). Jiang et al. (2024) used risk fields to represent off-road scenarios and designed global route planning and local trajectory planning algorithms. van der Ploeg et al. (2024) represented the impact of occlusions on driving via a risk field. While handcrafted potential field models can represent different traffic elements, such methods lack flexibility and generalizability when handling highly interactive scenarios.

Tree-based method: Due to its inherent reasoning capability, the tree-based autonomous driving planning method can explain the differences in the driving choices of different branches. Researchers have used this structure to explain various planning tasks. Omeiza et al. (2021) proposed a tree-based representation method to generate explanations for autonomous vehicle behaviors in different scenarios. User studies were conducted to evaluate the explanations. Gyevnar et al. (2022)

designed a planning process based on Monte Carlo trees and used Bayesian networks and text generation rules to provide causal natural language explanations for vehicle behaviors. Tree-based planning algorithms have good causal reasoning capabilities. However, few studies discuss how to use trees to generate visual explanations.

LLM/VLM-based: Recently, powerful base language models such as ChatGPT (Brown et al., 2020) and GPT-4 (Open et al., 2023) have gained widespread attention. Researchers are attempting to use LLMs or VLMs to understand traffic scenarios and explain autonomous vehicle behaviors. Dong et al. (2023) used fine-tuned LLMs to process driving videos and human queries, outputting the language model's understanding of driving scenarios. Yuan et al. (2024) proposed a retrieval-augmented generation (RAG) mechanism to compare similar driving scenarios, achieving better explainability and generalizability in understanding driving behaviors. While LLM/VLMs present new opportunities for explainable autonomous driving, their computational efficiency still needs improvement.

In this study, we integrate the first method with the second type of method above. Instead of potential field models, we apply a candidate trajectory tree to calculate dynamic spatial–temporal risks from predictions. Specifically, we study the collision risks from the ego vehicle's view and consider only those risks near the ego vehicle's reference path. In addition, considering that representing future driving risks on a single map cannot explain the risk changes over time, we use multiple risk maps to show the temporal evolution of future risks. A single risk map represents the spatial distribution of driving risk over a specific future period. In this study, we use three risk maps to show the future risk over the next 3 s. The first map shows the risk distribution from 0 to 1 s. These designs make the risk representation more intuitive and improve explainability. Furthermore, they enhance the computational efficiency of our method and its compatibility with mainstream perception outputs, such as occupancy grids and BEV maps. Considering these differences, we present this design as "spatiotemporal risk maps" in the following sections.

3. Method

In this section, we introduce our proposed method. Section 3.1 introduces the overall framework. Section 3.2 presents the detailed risk modeling algorithm. Section 3.3 presents the implementation details of trajectory planning.

3.1. Framework

As shown in Fig. 2, our method can be divided into three stages: scene prediction, risk modeling, and trajectory selection. The scenario prediction stage predicts the future trajectories of surrounding vehicles on the basis of the ego vehicle's perception data. In the planning stage, we use a candidate trajectory tree to model the dynamic driving risk from the prediction. The trajectory tree and corresponding risks are then projected to spatial–temporal risk maps. On the basis of the risk maps, we select the optimal trajectories from the tree according to the maximum acceptable cumulative risk and other driving objectives. In this work, we use the prediction network proposed by Geisslinger et al. (2021), which can estimate the uncertainty of the prediction while predicting the trajectories.

In the planning stage, we first model future driving risk. On the basis of a predefined driving behavior tree and the prediction outputs, we forward simulate potential interactions between the ego and the traffic environment, thus generating a candidate trajectory tree with risks. During the forward simulation, we convert the vehicle's behavior into a trajectory in the Frenet frame and then prune the trajectory branch and model its risk on the basis of the interaction results. Specifically, we

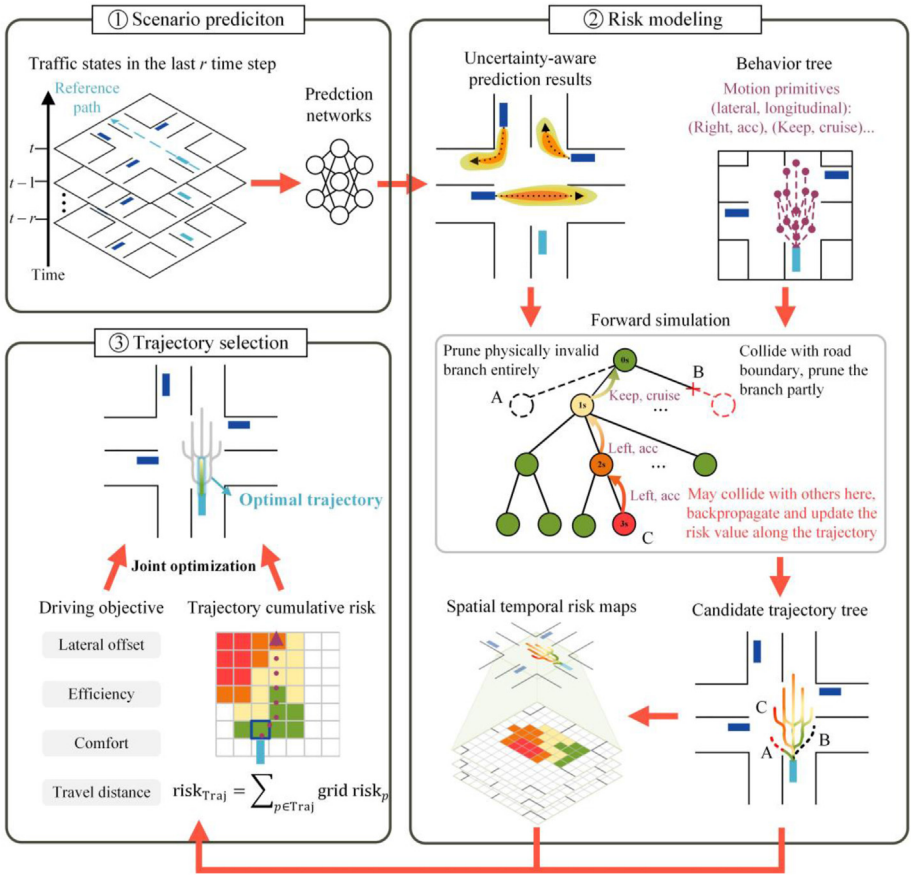

Fig. 2. Framework of the proposed method.

entirely prune trajectories that are physically infeasible (e.g., those with curvature or velocity that do not meet vehicle dynamics constraints) and partly prune trajectories that collide with road boundaries to ensure that the spatial risk distribution is within the drivable area. When a feasible trajectory may collide with other vehicles, we set the risk for that trajectory while considering safety factors such as collision points and velocity. The risk propagates along the parent branch, modeling the risk for the entire tree.

After risk modeling, we project the trajectory tree and its risks onto grid maps to form explainable spatial–temporal risk maps. Each level of the trajectory tree is projected onto an individual map. A trajectory is represented as an area on the grid map. The points in the trajectory are discretely mapped onto the grid and connected sequentially. The width of a trajectory area is the vehicle width, which reflects the range of the drivable area. If a trajectory collides with a vehicle, we consider the prediction uncertainty of the vehicle to determine the width of the area. When the uncertainty is high, the risk coverage area increases. Each time we draw a trajectory on the same grid cell, both the cell's risk and its drawing count accumulate. After plotting a level of trajectory branches, we divide the accumulated risk on the grid by the number of drawing counts, thus more fairly reflecting the future driving risk at that position.

Finally, we reevaluate all the candidate trajectories in the tree and select the optimal candidate. We perform discrete integration of the risk values at trajectory points to evaluate the expected risk of different driving behaviors. We use the median risk of all collision-free candidate trajectories as a threshold to filter out the lower-risk candidates. The filtered candidates are then evaluated in terms of driving efficiency and comfort. The candidate trajectory with the minimum cost is chosen as the final driving trajectory, achieving a balance between driving safety and efficiency.

Our algorithm is implemented via the CommonRoad framework (Althoff et al., 2017; Geisslinger et al., 2023; Maierhofer et al., 2021). Considering that this work focuses primarily on handling prediction in planning rather than trajectory prediction, we directly apply the prediction network and its pretrained weights proposed by Geisslinger et al. (2021). This network uses a CNN and LSTM-based encoder-decoder architecture. Other more advanced prediction network architectures (Hagedorn et al., 2024; Seff et al., 2023) could also be integrated into this framework to achieve more accurate risk modeling.

3.2. Risk modeling

To model the spatial–temporal risk of the ego vehicle's future driving, we first need to establish the relationship between the ego vehicle and its surrounding vehicles. We construct a three-level behavior tree in the Frenet frame to simulate common driving behaviors for the next 3 s. The branches have different lateral and longitudinal actions. The lateral actions include keeping the current offset, shifting left, and shifting right. The longitudinal actions include cruising (i.e., maintaining velocity), gentle acceleration, hard acceleration, gentle braking, and hard braking. Combining these primitive actions, we obtain 19 common driving behaviors within a 3 s duration (e.g., gently braking for 2 s to yield, then gently accelerating, accelerating and shifting left for 3 s to change lanes, increasing the braking force from gentle to hard, etc.). Notably, we set the forward simulation duration to 3 s since the algorithm is for local trajectory planning. However, our method can be extended to longer planning durations. For example, a four-level tree with each level representing 2 s is used to model risks over the next 8 s.

In the forward simulation, the predefined behavior tree is converted to a candidate trajectory tree on the basis of the current state of the vehicle. Fig. 3 illustrates this implementation process in a Frenet frame. The vehicle's position is quantified with two parameters: the lateral offset d and the longitudinal displacement s. In the behavior tree, each node includes two states: the target lateral offset d and the target longitudinal velocity v. A leaf node's state is determined by its parent and the primitive actions in the branch. The primitive actions are simple rules used to control the lateral offset and velocity, thus sampling the target states in the leaf nodes. We use the parent and leaf nodes as the initial state (d_0, v_0) and final state (d_f, v_f), respectively, and generate a trajectory curve in the Frenet frame (Werling et al., 2010). We employ suitable polynomial functions to generate lateral and longitudinal motion planning. Specifically, the trajectory along the lateral direction is represented as a quintic polynomial function of the form:

$$d(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5 \tag{1}$$

The polynomial coefficients $a_0, \ldots, a_5$ can be solved trivially by solving the following system of linear equations:

$$(a_0, a_1, a_2, a_3, a_4, a_5)^{\mathrm{T}} = E_{6\times 6}\left(d_0, d_f, \dot{d}_0, \dot{d}_f, \ddot{d}_0, \ddot{d}_f\right)^{\mathrm{T}} \tag{2}$$

where E refers to the functions of the initial and final lateral states. The final lateral state velocity $\dot{d}_f$ and acceleration $\ddot{d}_f$ are set to zero since the algorithm aims to plan a trajectory parallel to the reference path. For longitudinal motion planning, the quartic polynomial function is applied:

$$s(t) = b_0 + b_1 t + b_2 t^2 + b_3 t^3 + b_4 t^4 \tag{3}$$

The quartic polynomials in the longitudinal direction enable a manifold of the end positions since the velocity at the final state is not zero, whereas the acceleration is zero. Similar to Eq. (2), we can solve the coefficients $b_0, \ldots, b_4$ with the boundary conditions of the initial and final states (Werling et al., 2012). Each trajectory curve is 1 s long. To ensure

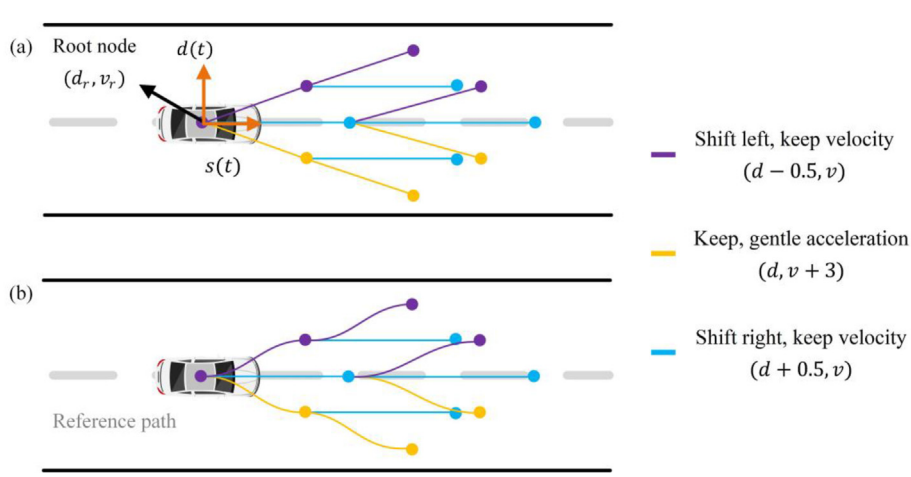

Fig. 3. Example to illustrate how the behavior tree is converted to a candidate trajectory tree in a Frenet frame: (a) behavior tree and (b) candidate trajectory tree.

the feasibility of generating trajectories, we check whether they meet vehicle dynamics constraints such as velocity and curvature limits. If not, the trajectory branch is pruned.

On the basis of the above trajectory generation method, we can build a candidate trajectory tree to model risks from interactions with obstacles in the environment. For static obstacles such as road boundaries, we directly prune trajectories that may collide with these obstacles, thus limiting the risk distribution within the drivable area. For dynamic obstacles, we calculate and propagate the risk on the basis of possible collisions with surrounding vehicles. We quantify the risk as a scalar value between 0 and 1. For a collision-free branch without any child nodes, we directly set its risk to 0. When a branch collides with a prediction trajectory, we set the risk for the collision point and its subsequent points to 1. For the points before the collision, we refer to the potential field (Wang et al., 2015) and set the risk in a gradually decreasing way:

$$r_t = r_{t+1} \bullet \max\left(0.8 - \frac{v_t}{20}, 0.2\right) \tag{4}$$

where r_t represents the risk of the trajectory point at time t, and v_t denotes the vehicle's velocity at time t. On the basis of the risk-setting method, a more conservative driving trajectory can have a lower risk when it approaches the collision point.

After calculating the risk of a trajectory branch, we need to back-propagate to identify the risk state of its parent branch. Fig. 4 shows an example for calculating the risk of a parent branch trajectory. We first calculate the average risk of all child trajectory branches:

$$r_{\text{child,avg}} = \frac{\sum_{t=t_1}^{t_m} r_t}{m} \tag{5}$$

where m is the number of points in a child branch trajectory. We set the average risk of these child branches as the risk of the final state in the parent branch:

$$r_{\text{final}} = \frac{\sum_{i=1}^{n} r_{\text{i,avg}}}{n} \tag{6}$$

where n is the number of child branches. For trajectory points before the final state, the risk values can be updated in reverse order via Eq. (4).

When performing risk modeling for a trajectory, we can plot it in the risk maps. Each level of the candidate trajectory tree is mapped into an individual risk map. There are three maps in total. Each risk map shows the driving risk around the reference path for different future periods. For example, the first level of the trajectory tree is mapped into the first risk map, which shows the risk distribution over the next 0–1 s. We grid the risk maps to improve computational efficiency. When plotting trajectories, we set different trajectory area widths w_{area} on the basis of the collision and prediction uncertainties. For collision-free trajectories, we directly set the width of the ego vehicle w_{ego} as w_{area}. If a trajectory branch collides with a predicted trajectory, we consider the impact of prediction uncertainty to set w_{area} as

$$w_{\text{area}} = w_{\text{ego}} + w_{\text{uncertainty}} \tag{7}$$

where $w_{\text{uncertainty}}$ represents the impact of increasing prediction uncertainty on risk coverage.

We set $w_{\text{uncertainty}}$ on the basis of the uncertainty of the collision point in the predicted trajectory. Here, we apply the error ellipse theory (Paradowski, 1997) to represent this uncertainty. Specifically, for the collision point (x,y), its uncertainty can be output by prediction networks as standard deviations σ_x and σ_y and covariance $\text{cov}(x,y)$. This uncertainty can be further expressed as an ellipse, where the axes S_x and S_y can be calculated as

$$S_x = \sqrt{1 + \frac{\text{cov}(x,y)}{\sigma_x \sigma_y}} \tag{8}$$

$$S_y = \sqrt{1 - \frac{\text{cov}(x,y)}{\sigma_x \sigma_y}} \tag{9}$$

with S_x and S_y, we can measure the coverage of uncertainty. Since we need to represent uncertainty as the trajectory width, we select the larger value between S_x and S_y as $w_{\text{uncertainty}}$.

$$w_{\text{uncertainty}} = \max\left(S_x, S_y\right) \tag{10}$$

In the above way, all trajectories are plotted onto the spatial–temporal risk maps. Multiple plots on the same grid cell accumulate risk and draw counts. Finally, we average the accumulated risk in each grid cell by its drawing count. It was plotted to measure the risk more objectively.

The above steps sequentially describe how we construct a candidate trajectory tree to model risk and then represent it in the risk maps. However, in practice, all calculations except averaging the grid cells can be executed in parallel during the construction of the trajectory tree. We use a recursive algorithm to achieve this implementation, avoiding repeated traversal on the tree. Algorithm 1 gives the detailed recursive implementation.

Algorithm 1. The recursive algorithm for risk modeling

Input: reference state, depth, behavior tree, current node, risk map
Output: the risk value of the reference state, valid trajectories T_V

```
Get the behaviors from the branches of the current node in the behavior tree.
Calculate the trajectories of the corresponding behaviors based on the reference state
Prune the physically invalid trajectories.
Check the collision of the trajectories and divide them into T_c (collide) and T_n (not collide).
  For each trajectory t_c in T_c:
    Set the risk of t_c and draw it in the corresponding map
  For each trajectory t_n in the T_n:
    If reaches the deepest level in the tree:
      Set the risk of t_n and draw it in the corresponding map
      Add t_n to the valid trajectories T_V.
    Else:
      Traverse to the next level with the final state of t_n as the reference state, get the returned risk value and the returned valid trajectories T_V'.
      Set the risk of the trajectory and draw the trajectory in the corresponding map
    For each trajectory t_v in T_V':
      Adds t_f to the front of t_v as t_v', then add to t_v' to T_V.
Calculate the average risk r_avg of the points in the trajectories
Return r_avg, T_V
```

3.3. Trajectory selection

After completing risk modeling, we obtain spatial–temporal risk maps and all valid collision-free trajectories. We first evaluate the safety risk of these candidate trajectories. However, directly using the risk modeled by the trajectory tree might not be comprehensive (i.e., the risk of a trajectory branch is only influenced by itself and its child branches, without considering other branches in the tree). Therefore, we evaluate the candidates' safety risk on the basis of the risk maps. A valid candidate trajectory consists of three trajectory branches. Each trajectory branch is projected onto a different risk map according to its time period. Fig. 5 shows an example, where both Traj A and the risk map are 2–3 s long. We map the trajectory points to the grid cells in the risk map. Then, we evaluate the risk level with the cells in which they are located. Finally, we calculate the cumulative risk of the candidate trajectory via discrete integration:

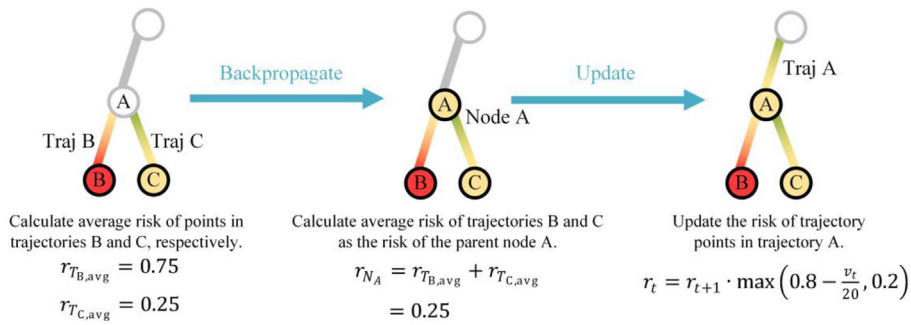

Fig. 4. Example to illustrate the backpropagation process. We update the risk of trajectory A according to the risk of trajectories B and C.

$$\text{risk}_{\text{Traj}} = \sum_{p \in \text{Traj}} \text{grid risk}_p \tag{11}$$

where p refers to the trajectory point. Since we have considered the risk decay with distance in risk modeling, we do not weigh the points differently but rather consider each trajectory point equally. We rank these candidate trajectories on the basis of their cumulative risk and select the half with the lowest risk.

We further evaluate the selected candidates while considering other driving objectives. Here, we use four common trajectory evaluation metrics:

1) Travel distance s: The distance traveled along the reference line in the Frenet frame.
2) Average lateral offset d_{avg}: The average lateral offset of points in the Frenet frame.
3) Average velocity v_{avg}: The average velocity of the trajectory points.
4) Energy consumption c: Energy consumption during the driving process, which can be measured as

$$c = \frac{1}{2}\int_{t_0}^{t_m} a^2 \mathrm{d}t \tag{12}$$

We calculate the above metrics for the candidate trajectories and standardize the results within each metric group. The cost function is a weighted combination of these metrics:

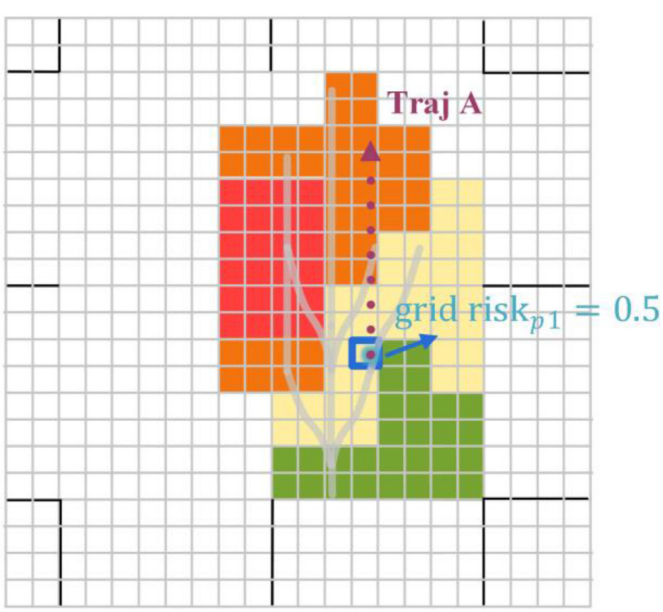

Fig. 5. Illustration used to calculate the cumulative risk of a trajectory point. We map trajectory point $p1$, onto the risk map and then evaluate the risk level using the grid cell in which it is located.

$$\text{cost} = w_1 s + w_2 d_{\text{avg}} + w_3 v_{\text{avg}} + w_4 c \tag{13}$$

where w represents the weight. We select the candidate trajectory with the minimum cost to balance driving safety, efficiency, and comfort.

4. Experiments

This section presents the simulation validation results of the proposed method. Section 4.1 introduces the simulation setup. Section 4.2 demonstrates the transformation from the candidate trajectory tree to the risk map. Section 4.3 analyzes the interpretability of the results. Section 4.4 presents the algorithm's runtime performance.

4.1. Simulation setup

We validate the proposed method on two datasets: Commonroad scenario (Althoff et al., 2017) and interaction (Zhan et al., 2019). We conduct open-loop experiments in the simulation. Specifically, we log-replay the dataset and let the vehicle follow the ground truth. In each frame, we compare the planned trajectories by our method against the ground truth from the human driver. This can be done with different metrics. As the name suggests, the planned trajectory will not be used to alter the state of the simulation. We applied the proposed method to all the vehicles in the datasets at each simulation step (0.1 s in this study). Here, we primarily present test results on several highly interactive traffic scenario datasets. Fig. 6 shows the maps and simulation information (i.e., simulation time and number of vehicles) for these scenario datasets. Notably, we keep the algorithm and the ego vehicle's parameter settings consistent across different scenarios to validate the algorithm's robustness. For the BEV input and the risk maps, we set the grid size to 0.5 m. Table 1 lists the target states corresponding to the primitive actions in the algorithm. Table 2 provides other parameter settings, including the algorithm's parameters for the ego vehicle and the weights in the cost function.

4.2. Case study: car following

We use a car-following example to illustrate the transformation from the candidate trajectory tree to the risk map. Fig. 7 visualizes the risk modeling process in a car-following scenario, where Vehicle 200 follows Vehicle 205 to leave the roundabout. We discuss the risk modeling process of the second level in the candidate trajectory tree, i.e., between the next 1 and 2 s The ego vehicle (Vehicle 200) is represented by a light blue dashed box located at the root node of the trajectory tree (i.e., the current position at 0 s). The dark blue Vehicle 205 is located at its position at 2 s. Fig. 7a presents the trajectory branches of the second level in the trajectory tree. The red trajectory indicates a rear-end collision occurring between 1 and 2 s. Fig. 7b visualizes the risk map between 1 and 2 s. The risk map is projected onto a grid. Different cells correspond to different risk values. There is a mapping relationship between the trajectory tree and the risk map. Both can reflect the increasing risk as the

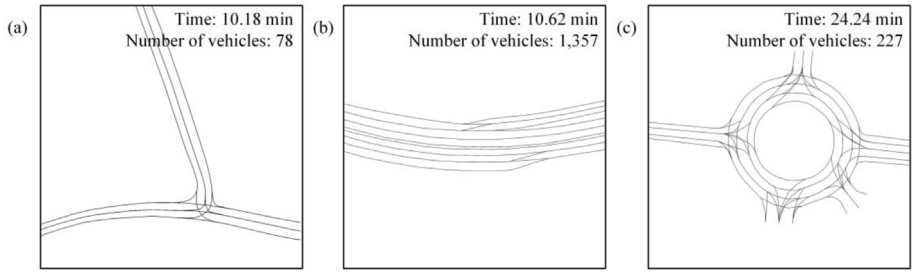

Fig. 6. Simulation scenario datasets and their information: (a) ZAM_Y-junction; (b) CHN_Merging; (c) CHN_Roundabout.

Table 1
Target state for executing primitive action (current state: lateral offset d_c, current velocity v_c).

Primitive action	Target state
Shift left	$d_c - 0.5$
Shift right	$d_c + 0.5$
Cruise	d_c
Gentle acceleration	$v_c + 3$
Hard acceleration	$v_c + 5$
Gentle brake	$v_c - 3$
Hard brake	$v_c - 10$

Table 2
Parameter settings in the algorithm.

Parameter	Value
Vehicle width	2 m
Max velocity	50.8 m/s
Max acceleration	11.5 m/s^2
Min steering angle	−1.066 rad
Max steering angle	1.066 rad
Min steering velocity	−0.4
Max steering velocity	0.4
Weight for travel distance	0.1
Weight for average lateral offset	−0.5
Weight for average velocity	0.1
Weight for energy consumption	−0.5

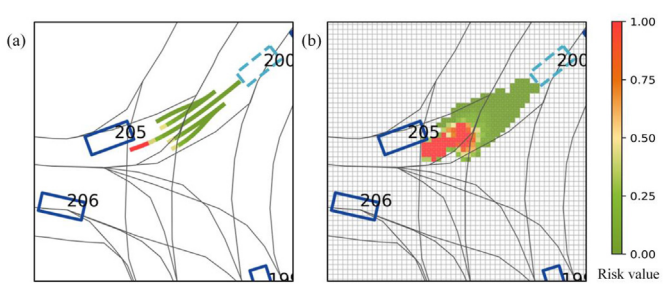

Fig. 7. Example to illustrate the relationship between the candidate trajectory tree and the risk map: (a) the second level of the trajectory tree; (b) the risk map during 1–2 s. The redder the color is, the higher the risk.

car-following distance decreases.

4.3. Evaluation of explainability

We further visualize risk maps for different scenarios to evaluate the interpretability of the method. Fig. 8 shows risk maps and trajectories for three different scenarios in the above datasets. The ego vehicle is the light blue dashed box. Other vehicles are in dark blue. In the leftmost diagram of each row, we visualize the initial state and trajectory prediction for the scenario. The following three diagrams are the risk maps for each second over the next 3 s. For clarity of presentation, we do not draw the grid here. Additionally, we visualize some candidate trajectories in the trajectory tree. For each scenario, we display the optimal trajectory and a collision-free suboptimal trajectory. In these diagrams, we anchor the ego vehicle's position at 0 s and then log-replay other vehicles' trajectories. In this way, we can evaluate the realism of the risk through evolution of the scenario.

Fig. 8a shows a T-junction scenario, where Vehicle 1 is moving straight and Vehicles 5 and 6 are turning left. The risk map for 0–1 s indicates that Vehicle 6 should drive conservatively, stop at the junction and wait. As Vehicle 1 leaves the T-junction, the risk map for 1–2 s shows a lower risk at the entrance of the T-junction. The risk map for 2–3 s suggests that Vehicle 6 should drive slowly when exiting the junction to avoid collision with the oncoming Vehicle 5. Comparing the optimal and suboptimal trajectories, although both trajectories yield Vehicle 1, the optimal trajectory slows down more gently. Thus, Vehicle 6 can pass through the T-junction more quickly.

Fig. 8b shows a merging scenario, where Vehicle 186 merges to the main road. The risk map for 0–1 s suggests that Vehicle 186 should merge and move forward. The risk maps from 1 to 3 s indicate that Vehicle 186 should follow the front vehicle at an appropriate velocity to prevent rear-end collisions with its front or rear vehicle. When the two trajectory candidates are compared, the suboptimal trajectory starts merging more slowly, increasing the risk of a rear-end collision with Vehicle 168.

Fig. 8c shows a roundabout scenario, where Vehicle 204 cuts in to leave the roundabout. The risk maps for 0–3 s indicate that Vehicle 211 should be cautious of the risk posed by Vehicle 204's cut-in. In addition, the risk maps for 1–3 s show that the risk on the left side of the reference path is significantly lower than that on the right side. This difference demonstrates that Vehicle 211 can shift left and then accelerate without increasing risk. Comparing the optimal and suboptimal trajectories, the optimal trajectory chooses the left path, allowing Vehicle 211 to pass quickly while maintaining a safe distance from Vehicle 204. In contrast, the suboptimal trajectory maintains a lower velocity, leading to lower efficiency.

4.4. Evaluation of trajectory planning

Following the settings in Section 4.1, we simulate 1,662 vehicles in three scenario datasets to evaluate the results of trajectory planning through open-loop experiments. For each evaluation, we compare the planned trajectories with the human driving trajectory over the next 3 s. We evaluate trajectories with the following metrics:

1) Average travel distance: This refers to the average distance traveled along the reference path over a 3 s trajectory. A longer distance indicates that the vehicle can reach the driving target more quickly.
2) Average velocity refers to the average velocity of all driving trajectories, which reflects the mobility of the traffic flow.
3) Average cumulative risk refers to the average risk of trajectories, which can be calculated with Eq. (11) in Section 3.3.
4) Average time headway: This term refers to the time difference between the ego vehicle and the leading vehicle passing the same point.

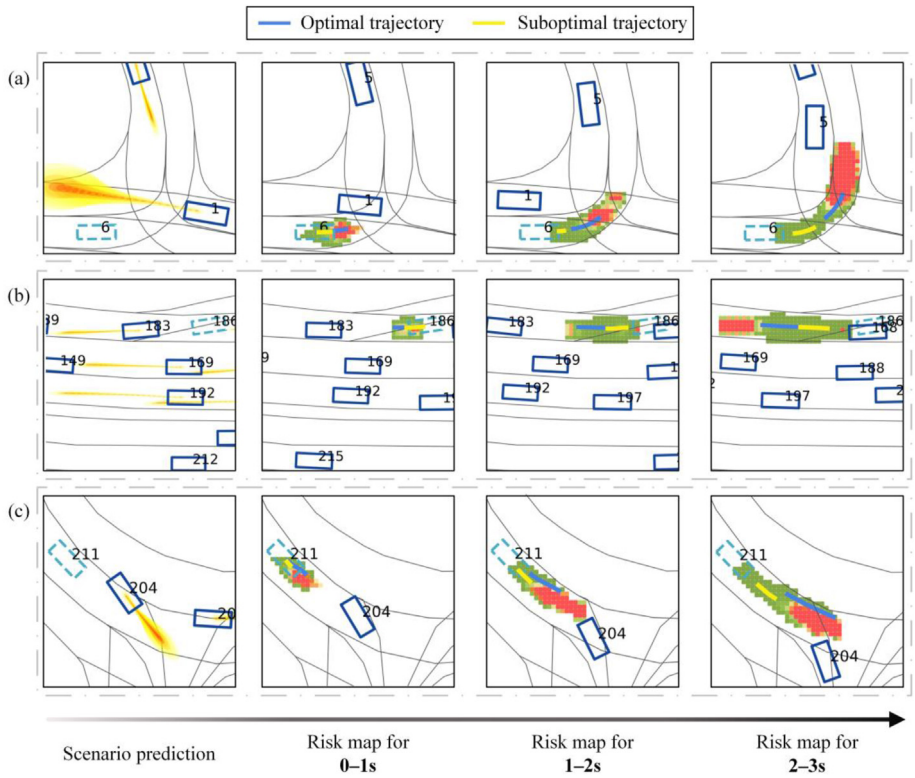

Fig. 8. Generated explanations in different scenarios. We visualize the spatial–temporal risk map and some candidates in the trajectory tree. For each row, the leftmost row shows the current scenario and predictions, while the remaining figure diagrams show the evolution of the risk map over time.

To compute the time headway, we project the leading vehicle at the current time step and let the ego vehicle move forward along the planned/driving trajectory. The time headway is defined as the first time the ego vehicle reaches the leading vehicle (up to 3 s). It can reflect the safety level of car-following behavior.

Table 3 presents the comparison results. Our method can achieve higher driving efficiency (i.e., longer distance and higher velocity) while accumulating lower driving risk. The results of the average headway indicate that our method can plan a safer trajectory for car following. Additionally, our method achieves lower risk and longer time headway in CHN_Merging, where car-following is the main driving behavior. This shows the consistency between our model's risk cognition and the classical metric.

4.5. Analysis of runtime

We further measured the runtime of different modules in our algorithm. Table 4 presents the average runtime for modeling risk and selecting trajectories in different scenario datasets. The results show that risk modeling requires more computational time than trajectory selection does. Additionally, there are differences in computation time across various scenarios. The computation time increases in scenarios with fewer vehicles (e.g., CHN_Roundabout). This is because fewer surrounding vehicles lead to less pruning, which increases the size of the candidate trajectory tree.

Although our code is implemented in Python and has not been optimized for performance, the average runtime is still within an acceptable range (100 ms). It can be expected that further optimization will improve its computational performance. This advantage can save onboard computational resources, achieving compatibility with other modules (e.g., a more complex and accurate trajectory prediction algorithm).

Table 3
Performance evaluation of the trajectory planning algorithm.

Scenario dataset	Method	Average travel distance (m)	Average velocity (m/s)	Average cumulative risk	Average time headway (s)
ZAM_T-Junction	Ours	14.53	4.99	6.29	1.86
	Dataset	11.34	3.91	9.96	1.65
CHN_Merging	Ours	14.35	4.94	17.93	2.38
	Dataset	12.68	4.38	21.60	2.19
CHN_Roundabout	Ours	13.79	4.75	6.33	2.02
	Dataset	12.52	4.32	11.15	2.04

Table 4
Runtime cost of different modules.

Scenario	Risk modeling (ms)	Trajectory selection (ms)
ZAM_T-Junction	66.3	2.4
CHN_Merging	64.8	2.2
CHN_Roundabout	76.0	2.8

5. Conclusions

In this study, we propose a planning framework that integrates spatial–temporal risk maps with a candidate trajectory tree to increase the interpretability of autonomous driving planning. We apply the candidate trajectory tree to achieve posterior risk modeling. This approach utilizes trajectory prediction for risk modeling and trajectory planning explicitly, providing better interpretability than end-to-end methods do. We further project the candidate trajectory tree onto the spatial–temporal risk maps. Thus, we can explain the relationship

between risks and different driving choices, offering better guidance for the safe driving of vehicles.

In the future, we plan to expand this work in the following directions. First, we intend to use ego-conditioned prediction to achieve more realistic interactions in forward simulation, thus modeling driving risk more accurately. Second, we plan to integrate our method with visual language models to generate language explanations for risk maps. Thus, we can use visual representations and natural language to better explain future driving risks.

CRediT authorship contribution statement

Qiyuan Liu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Conceptualization. **Jiawei Zhang:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Jingwei Ge:** Writing – review & editing, Visualization, Investigation. **Cheng Chang:** Visualization, Validation, Formal analysis, Data curation. **Zhiheng Li:** Writing – review & editing, Writing – original draft, Supervision, Investigation. **Shen Li:** Writing – review & editing, Software, Project administration, Methodology, Conceptualization. **Li Li:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization.

Replication and data sharing

The dataset and code used in this research is partially available upon request by emailing the corresponding author.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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